

Research Papers

Energy efficiency of lithium-ion batteries: Influential factors and long-term degradation

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ABSTRACT

As the integration of renewable energy sources into the grid intensifies, the efficiency of Battery Energy Storage Systems (BESSs), particularly the energy efficiency of the ubiquitous lithium-ion batteries they employ, is becoming a pivotal factor for energy storage management. This study delves into the exploration of energy efficiency as a measure of a battery's adeptness in energy conversion, defined by the ratio of energy output to input during the discharge and charge cycles. Energy efficiency values were systematically calculated over the course of the battery lifespan, revealing a predominantly linear trend in the efficiency trajectories, as substantiated by the Mann–Kendall (MK) trend test. Subsequently, a linear model was proposed to represent the efficiency degradation. Further analysis shows that ambient temperature, discharge current, and cutoff voltage all affect energy efficiency in different ways. Using the energy efficiency and its behavior observed in this study, Battery Management Systems (BMS) can improve the energy efficiency of batteries by adjusting operating conditions or developing better management strategies.

1. Introduction

Unlike traditional power plants, renewable energy from solar panels or wind turbines needs storage solutions, such as BESSs to become reliable energy sources and provide power on demand [1]. The lithium-ion battery, which is used as a promising component of BESS [2] that are intended to store and release energy, has a high energy density and a long energy cycle life [3]. The performance of lithium-ion batteries has a direct impact on both the BESS and renewable energy sources since a reliable and efficient power system must always match power generation and load [4].

However, battery's performance can be affected by a variety of operating conditions [5], and its performance continuously degrades during usage. The performance of a battery is related to various factors, irrespective of whether the equipment is in use or at rest. These determining factors include temperature, State of Charge (SOC), rest time, power rate, depth of discharge, and heat [6–8]. Each of these factors contributes to the overall performance and its degradation process, whether the battery is operational or static. As an energy storage device, much of the current research on lithium-ion batteries has been geared towards capacity management, charging rate, and cycle times [9]. A BMS of a BESS typically manages the lithium-ion batteries'

State of Health (SOH) and Remaining Useful Life (RUL) in terms of capacity (measured in ampere hour) [9]. As part of the management of BESS, it has been possible to estimate and predict SOH and RUL. SOH is a key indicator for the estimation of battery life by focusing primarily on the maximum capacity that the battery is currently capable to supply. SOC indicates the capacity that the battery currently can provide at the present time. Estimating the SOC can provide insight into the battery's current capacity, while the SOH trajectory can help predict the battery's life regarding its capacity. Despite the fact that the battery's capacity is one of the most critical performance indicators, limited attention has been devoted to understanding the factors influencing the energy efficiency of batteries and exploring this crucial aspect of battery performance.

The energy efficiency of lithium-ion batteries greatly affects the efficiency of BESSs, which should minimize energy loss during operations. This becomes increasingly important when more renewable energy sources are connected to the grid and handled by BESSs [2]. For these renewable energy sources to provide a stable, consistent power supply, it is essential that the batteries they rely on can deliver a high level of energy efficiency relative to the energy used to charge them. This energy efficiency is crucial not only for the performance of BESSs

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but also for maintaining the stability and reliability of the power grid as a whole. Hence, efforts should be exerted towards optimizing the energy efficiency of batteries, to ensure that they can meet the demands of a progressively renewable energy-dependent future.

Coulombic Efficiency (CE) [10] has been used as an indicator of lithium-ion battery efficiency in the reversibility of electrical current [11], which actually has a direct relationship with the battery's capacity [12]. It should be noted, however, that capacity and energy are not equivalent. Since the energy levels of lithium-ions are different during the redox reaction, regeneration requires more electromotive force than discharge due to the different voltage levels [13]. Therefore, even if lithium-ion battery has a high CE, it may not be energy efficient.

Energy efficiency, on the other hand, directly evaluates the ratio between the energy used during charging and the energy released during discharging, and is affected by various factors. For example, [14,15] examined how the cathode material affects a battery's energy efficiency. Several studies have calculated the one-way energy efficiency (energy efficiency in charging or discharging processes) of lithium-ion batteries and NiMH batteries under different charge and discharge rates [16,17]. [16] also compared the results between the two types of batteries. Some studies have also investigated the energy efficiency of batteries in different applications, such as electric vehicles [18], microgrids [19] and railways [20]. [20] used a BP neural network model to relate the state of charge, discharge rate and energy efficiency of titanate lithium-ion batteries. However, these studies did not consider the impact of aging on the battery's energy efficiency. [21] addressed this gap by examining the energy efficiency degradation of NMC and LFP batteries in calendar aging and used the Eyring Efficiency Fade Model and the Polynomial Efficiency Model to model the relationship between different SOC levels and energy efficiency. However, they only used data from the calendar aging of batteries and did not cover the energy efficiency degradation characteristics of lithium-ion batteries in cycling aging.

Since the energy efficiency of batteries is crucial for the planet's sustainability due to their wide range of applications [22], several studies have explored the use of BESS to achieve a satisfactory level of efficiency [23]. Some studies have also examined the energy efficiency of NiMH batteries in grid frequency control under different discharge currents [24]. To the best of the author's knowledge, no study has yet investigated what factors will, and how they affect the energy efficiency of lithium-ion batteries in the long run across their entire life cycle.

The objective of this study is to explore the trajectories in energy efficiency of lithium-ion batteries across their lifespan, specifically tracking the long-term degradation from initial deployment to their end-of-life (EoL). Furthermore, this study also delves into the impact of different operating conditions, including ambient temperature, discharge current, and cutoff voltage, on the battery's energy efficiency. The following are the contributions of this study:

- Energy efficiency in lithium-ion batteries is identified as a crucial metric, defined by the ratio of energy output to input during discharge and charge cycles.
- The degradation trajectory of energy efficiency for NCA lithium-ion batteries is studied and a linear model is proposed to describe energy efficiency degradation trend.
- A number of factors that affect energy efficiency have been identified and studied, including ambient temperature, discharge current, and cutoff voltage.
- We found NCA lithium-ion batteries have very weak memory effect on energy efficiency under changing operating conditions, which is helpful when designing an efficient BESS.

The remainder of this paper is organized as follows. In Section 2, the energy efficiency presented along with its relationship to CE and SOH, as well as the adopted dataset. In Section 3, the energy efficiency trajectories of batteries is analyzed in a variety of application scenarios. We verify the linear relationship between energy efficiency and cycle

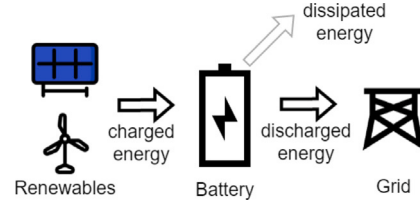


Fig. 1. Energy conservation during battery charging and discharging.

number by using time series analysis, and present the degradation trend model of battery energy efficiency and its regression results. Finally, in Section 4, we discuss the phenomena caused by changes in ambient temperature, discharge current, and cutoff voltage separately.

2. Energy efficiency of lithium-ion battery

2.1. Energy efficiency

As an energy intermediary, lithium-ion batteries are used to store and release electric energy. An example of this would be a battery that is used as an energy storage device for renewable energy. The battery receives electricity generated by solar or wind power production equipment. Whenever there is a demand from the grid, the stored electric energy is released. Inevitably, this process involves the dissipation of energy. As a result of polarization, the battery's energy dissipates during the charge–discharge process because coulomb losses from non-productive chemical side reactions and the battery's terminal voltage drops when current flows through it [25]. Therefore, while batteries are in operation, they lose energy during both cycle aging and calendar aging, and the amount varies depending on how they are used. Fig. 1 shows energy conservation in the application of battery. Energy dissipates during charging and discharging, but the energy in and out of the battery remains balanced.

$$E_{charge} = E_{discharge} + E_{dissipate}, \quad (1)$$

where $E_{discharge}$ and $E_{dissipate}$ represent the energy released during discharge and the energy dissipated during battery operation. The dissipated energy in operation can be expressed as

$$E_{dissipate} = E_{disCycle} + E_{disCalendar}, \quad (2)$$

where $E_{disCycle}$ and $E_{disCalendar}$ represent the dissipated energy during the cycle aging and calendar aging. Existing researches have studied SOH, which refers to the maximum remaining capacity of a battery over its rated capacity, that is, the capacity performance of a battery. In contrast to SOH, energy efficiency focuses on the battery's efficiency in using energy, as discharge energy in a battery is always less than charge energy.

The USA PNGV battery test manual [26] gives a intuitive definition of round-trip efficiency, but does not have a strict specific test protocol. PNGV round-trip efficiency is defined as

$$\text{Round-trip Efficiency} = \frac{\text{watt} \cdot \text{hours}(\text{discharge})}{\text{watt} \cdot \text{hours}(\text{regen})} \times 100\%. \quad (3)$$

Considering batteries are energy storage devices, we propose a calculation of battery efficiency based on the ratio of output energy to input energy. As this study aims to evaluate the energy efficiency of a complete charging and discharging process, energy efficiency is defined as

$$EE = \frac{E_{discharged}}{E_{charged}}, \quad (4)$$

where energy efficiency (EE) is calculated as the ratio between the amount of energy the battery can supply during discharge and the

amount of energy it consumes during charging. The charged energy is the accumulation of power during charging:

$$E_{\text{charged}} = \int_{t_0}^{t_n} P_{\text{charge}}(t) dt = \int_{t_0}^{t_n} V_{\text{charge}}(t) I_{\text{charge}}(t) dt, \quad (5)$$

where $P_{\text{charge}}(t)$ is the charging power as a function of time, which is obtained by multiplying $V_{\text{charge}}(t)$ and $I_{\text{charge}}(t)$, that is, the product of voltage and current at the terminal of the battery. It is common practice in industrial applications to sample discretely, then Eq. (5) can be simplified as

$$E_{\text{charged}} = \sum_{i=0}^{n-1} P_{\text{charge}} \Delta t_i = \sum_{i=0}^{n-1} V_{\text{charge}} I_{\text{charge}} \Delta t_i, \quad (6)$$

Similarly, we can also measure the energy during discharge. Hence, in this study, energy efficiency is evaluated as

$$EE = \frac{\sum_{i=0}^{n-1} V_{\text{discharge}} I_{\text{discharge}} \Delta t_i}{\sum_{i=0}^{n-1} V_{\text{charge}} I_{\text{charge}} \Delta t_i}. \quad (7)$$

where $V_{\text{discharge}}$, $I_{\text{discharge}}$, V_{charge} and I_{charge} are provided in the dataset and Δt_i is calculated by their timestamp.

2.2. Coulombic efficiency and energy efficiency

Coulombic efficiency or CE as a battery parameter to monitor the magnitude of side reactions [12] is defined as

$$CE = \frac{C_{\text{discharge}}}{C_{\text{charge}}}, \quad (8)$$

where $C_{\text{discharge}}$ is the discharge capacity of a battery at a single cycle, and C_{charge} is the charge capacity of the battery in the same cycle. By definition, CE is the ratio of discharge capacity over charge capacity of a lithium-ion battery. Because capacity is measured by the total charge flow to/from the electrode and the total capacity of lithium-ion batteries is usually cathode-limiting, CE can be expressed as the ratio between the amount of lithium-ions or electrons returning to cathode and the amount of lithium-ions or electrons departing from cathode in a full cycle, as indicated in Eq. (8).

A battery's CE refers to its ability to regenerate lithium-ion or electrons. In an ideal battery, where there are no side reactions at the electrodes, lithium-ions or electrons should flow solely as a result of reversible electrochemical reactions with CE equal to 100%. Realistic batteries, however, exhibit numerous side reactions between the electrode and the electrolyte. Due to the presence of irreversible side reactions in the battery, the CE is always less than 100%. Generally, modern lithium-ion batteries have a CE of at least 99.99% if more than 90% capacity retention is desired after 1000 cycles [11].

However, the coulombic efficiency of a battery cannot be equated with its energy efficiency. This is due to the fact that lithium atoms participate in redox reactions at different energy levels. The overpotential that is consumed during charging cannot be recovered during discharging. Due to the internal resistance of the battery, some energy must be consumed to overcome the resistance, which limits the battery's ability to achieve 100% energy efficiency [13].

Fig. 2 illustrates theoretical scenarios that identical CE values could correspond to significant differences in energy efficiency due to potential variations. These illustrations serve to underscore the distinction between CE and energy efficiency, especially in the context of energy conversion efficiency in battery energy storage applications. More specifically, for the ideal 100% energy efficiency in (a), the charge/discharge curves are perfectly symmetrical, meaning that the stored lithium-ions have the same energy level as in both the charge and discharge phases. Nevertheless, the electrons of the charge phase have a higher energy level than the electrons of the discharge phase in (b), which will result in an energy efficiency of only 90%. (c) represents a very low energy conversion efficiency of the battery, where 50% of the energy is wasted in the charging/discharging process despite the fact that 100% of capacity is available.

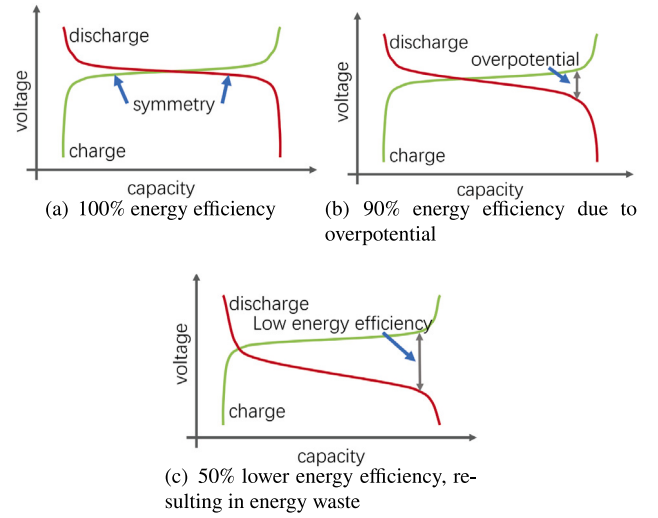


Fig. 2. Three different scenarios [13] at 100% CE.

Table 1

Technical indicators of experimental lithium-ion batteries.

Manufacturer	Idaho National Laboratory
Type	18650
Cathode/Anode	NCA/Graphite
Rated capacity/Rated voltage	2 Ah/3.7 V
Charging currents	1 A for CC stage
Cutoff voltages	4.2 V to 2/2.2/2.5/2.7 V
Discharging currents	1/2/4 A
Ambient temperature	4/24/43 °C
Stop criteria	20/30% capacity fade

2.3. Public lithium-ion battery datasets

We need data over the entire lifespan of lithium-ion batteries in order to model the degradation of energy efficiency, and to analyze what factors affect the energy efficiency of these batteries. Publicly available lithium-ion battery datasets [27–31] are not all derived from round trip cycle aging experiments. Moreover, few are conducted under varying operating conditions (ambient temperature, charging current, discharging current, and depth of discharge) that influence battery performance.

The dataset, which is provided by the Prognostics Center of Excellence (PCoE) at NASA Ames [32], is derived from experiments [33] in which 18650-size Nickel-Cobalt-Aluminum (NCA) lithium-ion batteries continue to work until EoL is reached. This is helpful to see how energy efficiency has changed over time.

These experiments, as shown in Table 1, are designed to reduce battery performance in various controlled ways. The test scheme, as shown in Table 2, is with a test matrix that included four cutoff voltages (2.7 V, 2.5 V, 2.2 V and 2.5 V), three discharge currents (1 A, 2 A, and 4 A), four ambient temperatures (4 °C, 24 °C, 43 °C), and two stopping criteria (20% and 30% capacity fade). In most cases, an individual battery's operating conditions are different from those of other batteries, and remain constant throughout the test. The dataset has gathered experimental data on the aging of NCA lithium-ion batteries under a wide range of operating conditions. Although the test cases of this dataset are far from thorough, it still provides interesting observations for an initial study on the affecting factors of battery energy efficiency.

Each cycle, batteries are fully charged in CC-CV (Constant Current, Constant Voltage) mode and discharged in CC (Constant Current) mode to a certain depth, which is regulated by cutoff voltage. Throughout the cycle, voltage and current at the battery terminal are collected, which can be used to calculate how efficiently energy is reproduced in the specific cycle.

Table 2
Test scheme of batteries.

Ambient Temperature	4 °C		24 °C		43 °C	24 °C 4 A ⇒ 44 °C 1 A ⇒ 44 °C 2 A
Discharge current	1 A	2 A	2 A	4 A	4 A	
cutoff voltage						
2.0 V	B0045	B0053		B0033	B0029	
2.2 V	B0046	B0054	B0007	B0034	B0030	B0038
2.5 V	B0047	B0055	B0006		B0031	B0039
2.7 V	B0048	B0056	B0005		B0032	B0040

The dataset also includes 3 batteries that undergo 3 different discharging conditions during the tests: first at 24 °C 4 A, then at 44 °C 1 A, and finally at 44 °C 2 A until they reached EoL. This specific group of batteries can offer information about how altering operating conditions influences energy efficiency for the same battery.

According to some studies, a fresh battery's CE may exceed 100% because the internal electrochemical characteristics are not stable in the initial cycles [11]. Furthermore, the documentation of the dataset indicates that there are abnormalities due to software errors or other unknown causes, such as stopping the test before reaching the specified cutoff voltage. Considering that the purpose of this study is to determine the characteristics of lithium-ion battery's energy efficiency throughout its lifespan, as well as to identify the trend of energy efficiency and the influence of relevant operating conditions on it, the first cycle as well as some special abnormal cycles for each battery have been excluded.

2.4. SOH and energy efficiency

Numerous studies have been conducted on SOH for energy storage batteries. Served as a measure of the battery's aging process, SOH is commonly calculated as

$$SOH = \frac{Q_{max}}{Q_{rated}}, \quad (9)$$

where Q_{max} and Q_{rated} represent the battery's maximum discharge capacity with its rated capacity. RUL is defined by the number of cycles a battery can undergo before its capacity deteriorates and reaches EoL which is typically defined as when the battery has only 80% of its rated capacity left.

Fig. 3 illustrates typical energy efficiency and SOH trends as the battery ages over a number of cycles. It should be noted that the SOH trajectories in this dataset did not show noticeable 'knee-points' that existed in some other datasets [30] where the rate of degradation accelerates significantly, which could be the results of differences in experimental settings or the specific type of batteries under study.

But nonetheless from Fig. 3 we can see that, due to the wide range of use scenarios included in the dataset, including different discharge intensities and depths in various ambient temperatures, these batteries exhibit distinctly different trajectories in terms of their SOH and energy efficiency. As an example, when the battery B0005 is tested at an ambient temperature of 24 °C, with a discharge current of 2 A and a cutoff voltage of 2.7 V, its energy efficiency is essentially above 0.83 while its SOH is continuously reduced to less than 0.70. As a comparison, B0033 was also evaluated at 24 °C, but is subjected to a higher intensity (4 A current) and a deeper (2.0 V cutoff voltage) discharge, its energy efficiency is around 0.7. The efficiency of B0005 was 0.86 when its capacity had decayed to about 0.8, whereas B0033's energy efficiency was only 0.73 with similar SOH. In spite of the high discharge temperature of 43 °C, the SOH of B0029 drops very rapidly, but it is almost always above 0.83 in terms of energy efficiency. One of the extreme cases was the testing of B0045 at 4 °C. Even though its SOH was very low (less than 0.5 throughout the test), it had a relatively high efficiency of about 0.78. As illustrated in these examples, SOH and energy efficiency differ in how they explain battery performance.

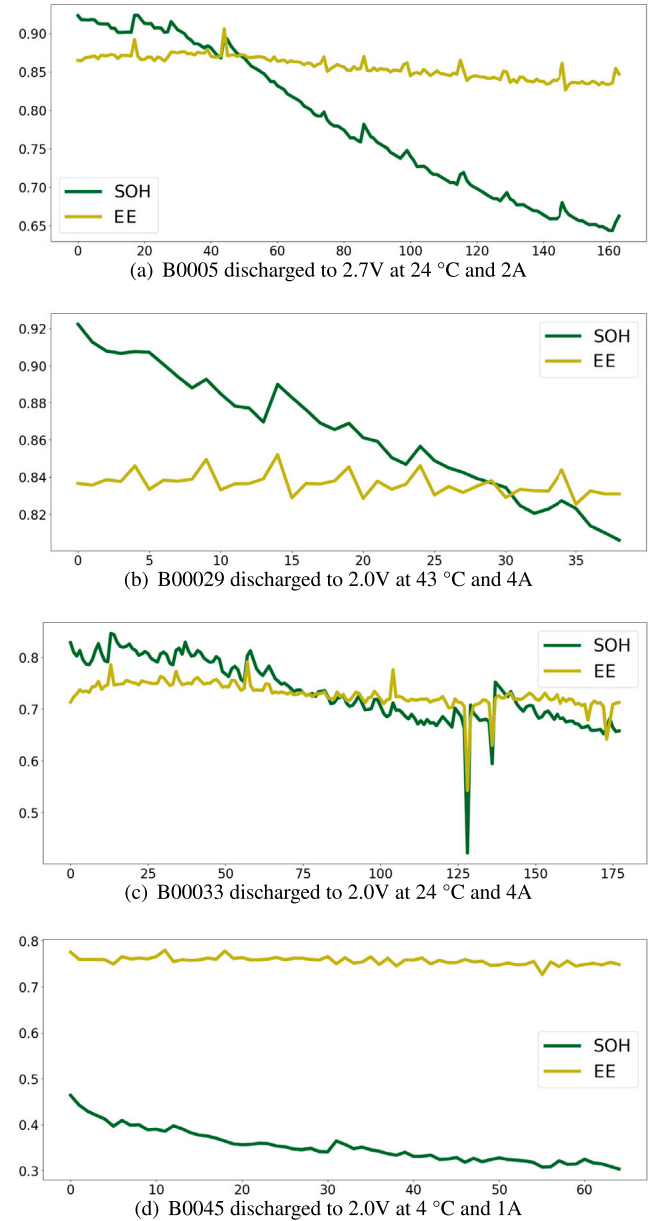


Fig. 3. Typical SOH and energy efficiency trajectories over cycling.

Generally, SOH describes the health of a battery in terms of its ability to release coulombs. While energy efficiency describes the efficiency of a battery as an energy storage medium in terms of the ratio of energy transfer during charging and discharging. Further details on typical energy efficiency and SOH values can be found in Table 3.

Table 3
Typical energy efficiency and SOH values.

Batteries	SOH	Energy efficiency
B0005	90.23%	87.59%
B0039	88.58%	90.50%
B0005	80.08%	86.10%
B0033	81.28%	73.73%
B0005	70.06%	84.45%
B0045	46.40%	77.55%

To quantitatively analyze the correlation between energy efficiency and SOH, we used the person correlation coefficient (PCC). the PCC is calculated as follows:

$$PCC = \frac{\sum (EE - \overline{EE})(SOH - \overline{SOH})}{\sqrt{\sum (EE - \overline{EE})^2 \cdot \sum (SOH - \overline{SOH})^2}} \quad (10)$$

As shown in the Table 4, the number above each battery label represents the PCC value. Lighter background color indicates smaller PCC value and weaker correlation. Notably, there is not a clear relation between SOH and energy efficiency for all batteries, and evaluating specifically for the efficiency of batteries remains a necessity.

3. Modeling energy efficiency

3.1. Linear trend of energy efficiency trajectory

A battery undergoes a series of charging and discharging cycles during its aging process. For the dataset used, the SOH of each battery decreases over time until it reaches the end-of-experiment condition (typically 20% or 30% capacity fade). According to Eq. (7), we calculate the energy efficiency for each battery in each of its charging/discharging cycle. Fig. 4 shows the trajectory of energy efficiency (ranging from 0 to 1) across the cycles a battery undergoes before reaching its EoL. Despite these batteries reaching the EoL level of capacity fade under specific and even harsh conditions, the decline in their energy efficiency is not as pronounced. Furthermore, batteries exhibit different trajectories of energy efficiency degradation under varied operating conditions.

While experiencing substantial fluctuations in their energy efficiency trajectories, all these batteries display an overall flat or slightly downward trend in energy efficiency. This implies an observed linear trend.

Now we want to test and prove that the energy efficiency trajectory is indeed linear. For this purpose, we first calculate the first difference of the energy efficiency time series, which removes the first-order trend from the original trajectory; if the resulting first difference has not trend any more, then we can say that the original trajectory has a linear trend. The first difference for EE_t is expressed as

$$\Delta_{EE_t} = EE_t - EE_{t-1} \quad (11)$$

To determine whether the Δ_{EE} shows no trend, we use The Mann-Kendall (MK) Trend Test. The MK Trend Test [34,35] is used to analyze time series data for consistently increasing or decreasing trends. MK trend test is also used in trend detection in hydrologic data [36]. For the time series Δ_{EE} with the actual number of cycles performed in the experiment of n (sample size), the MK trend test first calculates S :

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(\Delta_{EE_j} - \Delta_{EE_k}), \quad (12)$$

$$\text{where } \text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x = 0 \\ -1 & \text{if } x < 0 \end{cases}.$$

An MK test analyzes the sign of the difference between all data and its predecessors. Throughout the time series, each value is compared

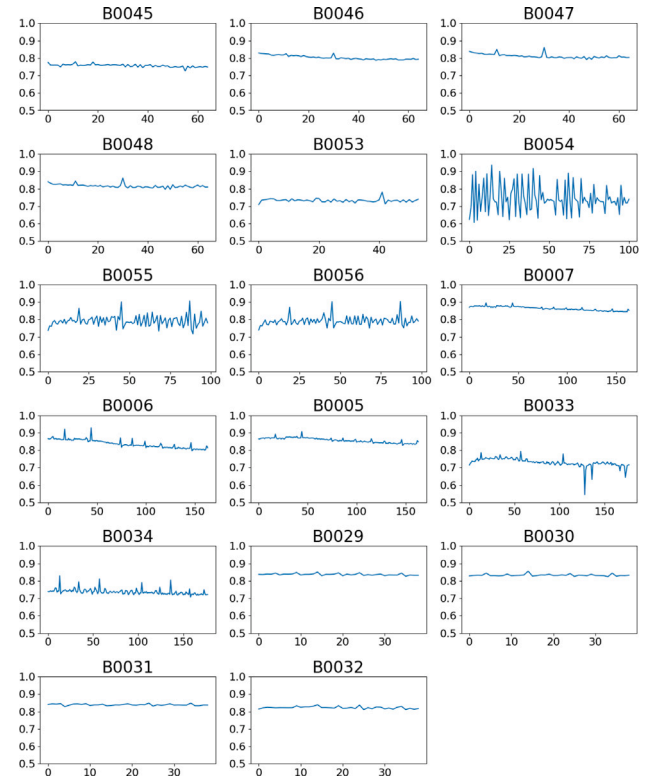


Fig. 4. Energy efficiency trajectories.

with its preceding value, giving $n(n-1)/2$ pairs of sign values. Intuitively, the number of sign values indicates the tendency of a trend's present. The variance of S is then calculated as

$$\text{VAR}(S) = \frac{n(n-1)(2n+5) - \sum_{k=1}^g q_k(q_k-1)(2q_k+5)}{18}, \quad (13)$$

where g represents the number of tie groups of the time series energy efficiency, and q_k represents the number of data of k tie groups. For the test statistic, transform S into Z_{MK} as follows:

$$Z_{MK} = \begin{cases} \frac{S-1}{\sqrt{\text{VAR}(S)}} & \text{if } S > 0 \\ 0 & \text{if } S = 0 \\ \frac{S+1}{\sqrt{\text{VAR}(S)}} & \text{if } S < 0 \end{cases} \quad (14)$$

Using the python package [37] of MK trend test, the Δ_{EE} series of each battery were examined and the results are shown in Table 5. The z -table can be used to calculate the probability (p -value).

After computing Z_{MK} , a p -value can be determined as:

$$p = 2(1 - \Phi(|Z|)), \quad (15)$$

where Φ represents the cumulative distribution function of the standard normal distribution. Typically, if the p -value is less than 0.05, it indicates a trend in the time series. Given that the p -values for the batteries detailed in Table 5 all surpass 0.10, we confirm that there is no consistently increasing or decreasing trend in the first difference of the energy efficiency series. Therefore, the energy efficiency trend of these batteries should obey a linear trend.

3.2. Model and regression results

Building upon the previous discussions which revealed a predominantly linear trend in the energy efficiency trajectories of the NCA lithium-ion batteries throughout their lifetimes, we are motivated to develop a model that reflects this observation. Accordingly, we propose

Table 4
PCC of SOH and energy efficiency.

0.5406 (B0045)	0.9218 (B0046)	0.7705 (B0047)	0.6512 (B0048)	0.0333 (B0053)
0.0502 (B0054)	0.0254 (B0055)	0.1265 (B0056)	0.9442 (B0007)	0.9161 (B0006)
0.8326 (B0005)	0.9183 (B0033)	0.6205 (B0034)	0.4282 (B0029)	0.1962 (B0030)
0.1075 (B0031)	0.3998 (B0032)			

Table 5
The p-values of Mann-Kendall trend test result.

B0045	B0046	B0047	B0048	B0053
0.767631	0.434133	0.689333	0.655519	0.957178
B0054	B0055	B0056	B0005	B0006
0.700847	0.882874	0.662952	0.870048	0.579624
B0007	B0033	B0034	B0029	B0030
0.774103	0.998988	0.698799	0.919888	0.687463
B0031	B0032	—	—	—
0.782101	0.880084	—	—	—

a linear model for energy efficiency, which can be represented by the following equation:

$$EE_t = \alpha t + \eta + v_t, \quad (16)$$

where α is the slope of the trend, t is the cycle number, η is the energy efficiency of the battery at the beginning of the test, and v_t is the fluctuation around the linear trend.

As the fluctuations, which arise from stochasticity, are not dependent on the trend, we can represent the trend of energy efficiency of a specific NCA lithium-ion battery under a certain operating condition with two parameters: the slope α and the initial value (the intercept) η .

$$EE_t = \alpha t + \eta. \quad (17)$$

Batteries in the dataset that have been aged under various operating conditions. We use least squares regression to fit the energy efficiency trajectories of every battery using energy efficiency model presented as Eq. (17). A selection of regression results are shown in Fig. 5.

As the operating conditions of these batteries differ mainly in terms of ambient temperature, discharge current, and cutoff voltage. To collectively visualize the effect of different operating conditions on the energy efficiency of the battery, a summary of the results is shown in Fig. 6.

As shown in Fig. 6(a), a straight line represents a specific operating condition during the battery life cycle. Longer lines indicate a longer battery life. The blue, green, and red lines correspond to ambient temperatures of 4 °C, 24 °C, and 43 °C respectively. A thinner line indicates a lower discharge current, whereas a thicker line indicates a higher discharge current, and a thick line with edge indicates discharge current of 4 A. Dark colors represent deep cutoff voltages, and light colors are for high cutoff voltages.

Fig. 6(b) shows the energy efficiency ranges of the batteries. Each horizontal bar represents an energy efficiency span from a particular test case. Right-handed bars have a higher energy efficiency, whereas left-handed bars have a lower energy efficiency. At the beginning of the test, the energy efficiency is located on the right side of the bar. As the battery ages, the energy efficiency shifts to the left. Longer bars indicate a battery with a wider range of energy efficiency during the test, which means these batteries show a more obvious decline in energy efficiency during the test.

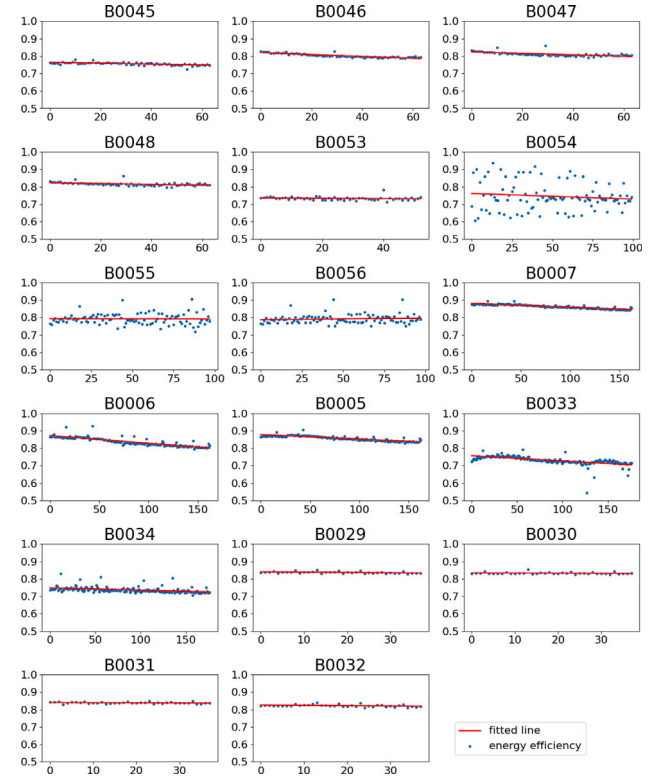


Fig. 5. Energy efficiency regression results.

4. Analysis and discussion

4.1. Energy efficiency trends and ranges under different operating conditions

The test schema specifies that EoL conditions occur when battery capacity drops below a certain level, at which point the test is terminated. As a result, the batteries undergo a variation in the number of cycles. Depending on the rate at which battery capacity is degraded, energy efficiency trajectories vary in length. Among all the constant operating conditions included in the dataset, batteries has the longest life at 24 °C, followed by 4 °C, and the shortest at 43 °C. However, contrary to SOH, the batteries with the highest energy efficiency were tested at 43 °C, 1 A. This suggests that discharge current may have a greater effect on energy efficiency than ambient temperature.

Continuous cycling decreases the energy efficiency of most batteries. Batteries, which is discharged at 4 °C 2 A current, despite having a relatively long RUL, had a relatively lower energy efficiency than those discharged at 4 °C 1 A. It is possible that the higher discharge current may have contributed to an extended RUL, but resulted in a suppressed energy efficiency for batteries at extra low temperatures. It is interesting to note that these batteries suffer severely from lower cutoff voltages in terms of energy efficiency at 4 °C ambient temperature. The energy efficiency of batteries discharged at 4 °C 1 A with

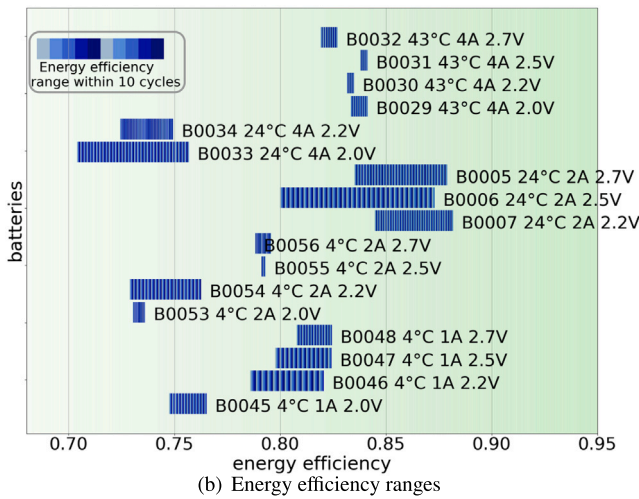
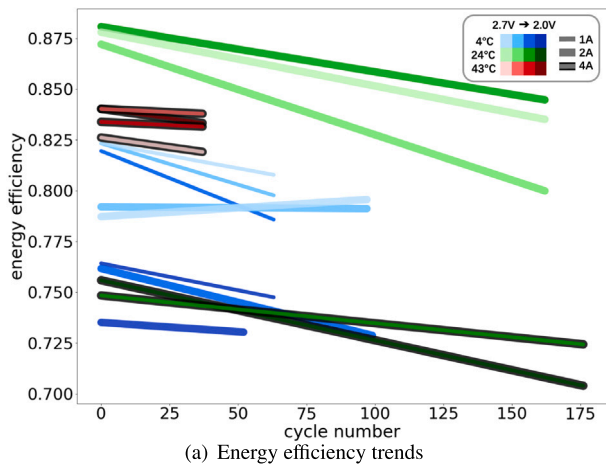


Fig. 6. Energy efficiency trends and ranges under various operating conditions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

a voltage of 2.0 V and 2.2 V has a value of approximately 0.75, while other batteries of the same group with a relatively higher cutoff voltage, have a value of approximately 0.8.

Batteries cycled at higher temperature undergo rapid degradation in terms of SOH, yet their energy efficiency appears to be positively impacted. For instance, consider B0030 and B0034, which both operate with identical discharge currents and cutoff voltages but under different ambient temperatures; B0030 at 43 °C and B0034 at 24 °C. When it comes to battery longevity in terms of cycles, B0030 falls short compared to B0034. However, B0030 maintains a higher energy efficiency throughout its lifespan than B0034.

When tested at 24 °C with a 2 A discharge current, batteries exhibit a long RUL and a high energy efficiency. In these batteries, the cutoff voltage appears to have a mitigating effect on energy efficiency, and RUL and energy efficiency may be affected by differences in the manufacturing process. When the batteries were tested with 4 A at the same temperature, their energy efficiency was again suppressed by the discharge current. It is noteworthy that the dataset does not include the 24 °C 1 A test case. Based on all the results in the dataset, it appears that this approach will result in higher efficiency and RUL.

The range of energy efficiency for these batteries is determined by the slope of the trend and the number of cycles. Batteries that have a relatively long RUL and a high tendency to degrade have a longer energy efficiency range. Batteries operating at 24 °C 2 A have a high initial energy efficiency and a wide energy efficiency range. These

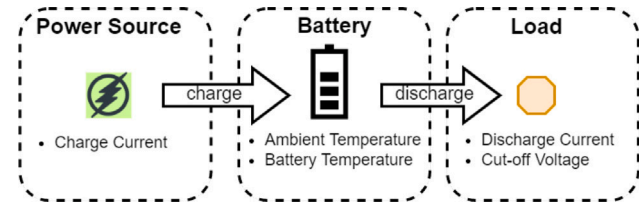


Fig. 7. The influence factors on battery energy efficiency.

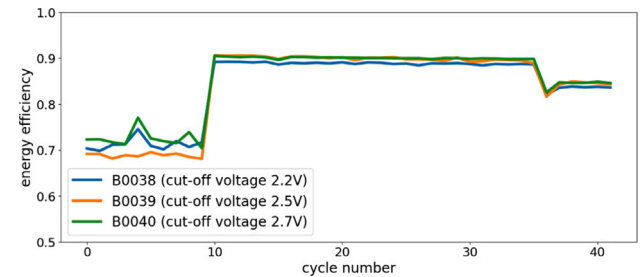


Fig. 8. Energy efficiency curves for a special group with varying operating conditions.

characteristics indicate that the batteries' energy efficiency is relatively good at the beginning of the test and decreases as they age. In contrast, batteries that operate at 43 °C have a narrow energy efficiency range, primarily due to their short RUL. 4 °C 2 A batteries have little tendency to decrease with a higher cutoff voltage, so they also have a narrow energy efficiency range.

4.2. Influence of operating conditions

Managing the energy efficiency of lithium-ion batteries requires optimization across a variety of factors such as operating conditions, charge protocols, storage conditions, and Battery Management System (BMS) regulations. Among these factors, the operating conditions exert a substantial influence on battery efficiency, dictating the battery's performance under different usage intensities and varying temperatures, as shown in Fig. 7. We use NASA's battery dataset, which utilizes a variety of combinations of operating conditions across distinct battery groups, to investigate their performances under diverse circumstances.

Moreover, the B0038, B0039, and B0040 batteries had test conditions that changed during the aging period as opposed to the other groups with constant operating conditions. These batteries began their cycle at an ambient temperature of 24 °C with a discharge current of 4 A. After the first 11 cycles, the temperature was increased to 43 °C, and the discharge current was adjusted to 1 A. Later, the current was further increased to 2 A at the same temperature. Therefore, the operation of this battery group can be segmented into three distinct periods: 24 °C at 4 A, 43 °C at 1 A, and 43 °C at 2 A. As depicted in Fig. 8, when the operating conditions change, the energy efficiency of these batteries changes rapidly. However, under stable operating conditions, energy efficiency exhibits stable linear characteristics, this may indicate that there is slight memory effect in the energy efficiency of the batteries when the operating conditions change.

Fig. 9(a) shows that a battery with a lower discharge current is more energy efficient. Higher discharge currents allow a battery to operate at higher power, but they may also negatively affect the battery's energy efficiency. A B0034 discharged at 4 A has a energy efficiency of roughly 0.73. On the other hand, the B0007 discharged at 2 A has an energy efficiency of more than 0.85, at the same ambient temperature and cutoff voltage.

Battery performance increases at higher ambient temperatures, as shown in Fig. 9(b). From 4 °C to 24 °C, the energy efficiency of a 2 A discharged battery improved by almost 0.12; from 24 °C to

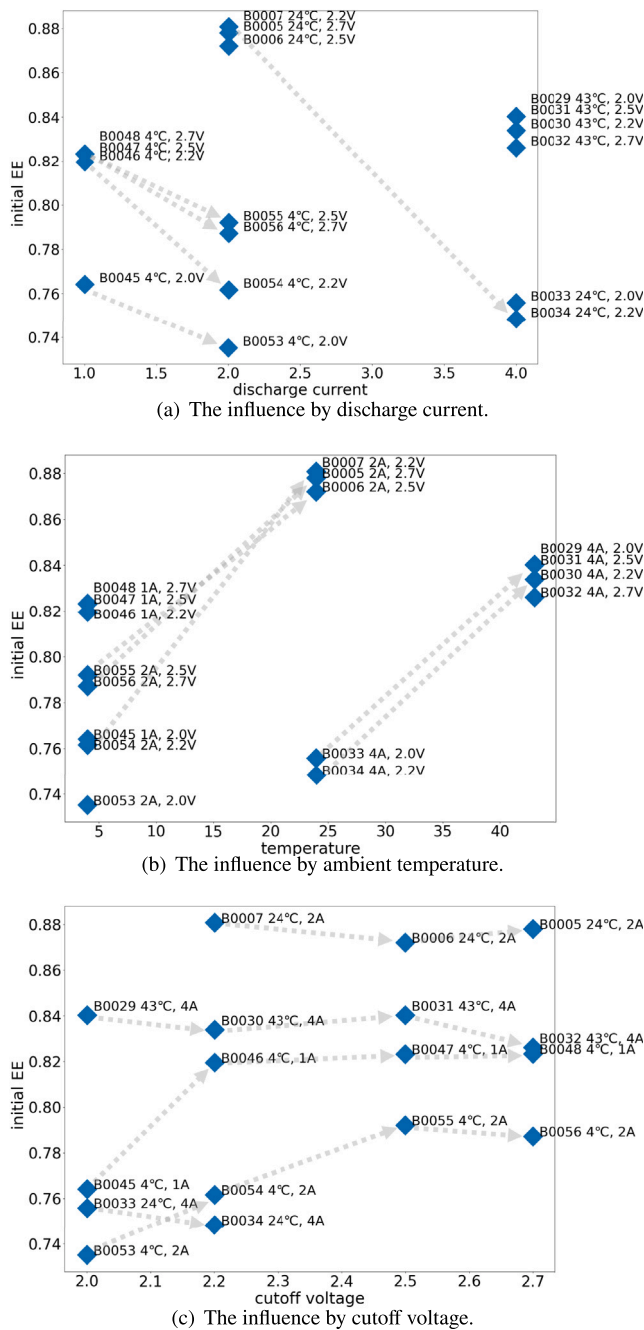


Fig. 9. The influence to energy efficiency by operating conditions.

43 °C, the energy efficiency of a 4 A discharged battery improved by approximately 0.09.

Discharging batteries at different depths can be achieved by using different cutoff voltages. When a battery is discharged to an extended depth, more energy is released during a single discharge cycle. An increase or decrease in discharge depth, for example, from 2.7 V to 2.5 V, generally has a limited effect on the energy efficiency, as shown in Fig. 9(c). It should be noted, however, that at ambient temperatures of 4 degrees or 24 degrees, 2.2 V or 2.0 V cutoff voltages will result in a significantly lower energy efficiency than those at higher cutoff voltages. Using the B0029 as an example, at a depth of discharge of 2.0 V, the energy efficiency is only 0.76; other batteries in the group, the B0032 and B0032, with cutoff voltages of 2.5 V and 2.7 V, have energy efficiencies exceeding 0.86.

4.3. Implications and suggestions

Based on NASA's experimental data, we identified some interesting observations and patterns concerning battery energy efficiency. Some of them make sense intuitively, while some are somewhat surprising, nonetheless in many cases they are quite different from the currently existing degradation patterns for SOH or SOC. Clearly, it would be worthwhile to delve further into the reasons underneath these findings, but these observations and patterns themselves can already serve as valuable references when it comes to using these batteries and designing efficient-aware BESSs in the future.

As expected, energy efficiency also gradually degrades just like SOH as the battery ages, but at a much slower pace: when operated under the same conditions, the overall energy efficiency degradation is rarely greater than 4% throughout its lifespan when its SOH reaches the 70% stopping point. In comparison, changes in operating conditions on the other hand have a much larger impact on energy efficiency that can easily result in more than 20% difference in energy efficiency. What is more, energy efficiency does not seem to exhibit a significant 'memory effect': when the operating conditions of a battery changes, in the following cycles its energy efficiency will change immediately and reaches a comparable level as if it has been operated under those conditions all along, irrespective of its previous operating conditions.

These behaviors of energy efficiency suggested that, old batteries that are currently considered unusable due to capacity loss, may actually still be useful efficiency-wise, since under the favorable operating conditions, their energy efficiency will not be much different from that of new batteries. There is still considerable potential for these batteries to provide efficient energy caching for renewable energy, and capacity loss can always be compensated by quantity. In this way, retired batteries can largely be reused for storage purposes such as caching the renewable energy in a cost-effective, more environmentally friendly and efficient manner.

Comparing to aging, operating conditions are much larger factors that determine the energy efficiency of a battery. For BESS, the performance of batteries varies due to production deviations, inhomogeneous aging, and temperature differences within the cluster. With a battery pack containing batteries with differing energy efficiencies and SOHs, the overall energy efficiency of BESS may be adversely affected. In order to maximize the performance and efficiency of a battery pack, active balancing must be implemented in order to equalize the performance of these batteries.

First, intensive discharging at high current always has a negative impact on energy efficiency, therefore to improve the overall efficiency of a BESS, we should carefully control the discharge current of individual batteries and keep them balanced and at a lower level. On the other hand, although it has been well received that both low and high temperatures have a detrimental effect on battery capacity or SOH, the effect of higher ambient temperatures on energy efficiency, however, is essentially positive. In other words, the battery's energy efficiency can be improved when the temperature is kept warm within its normal operating range. Low temperatures have a further negative effect on energy efficiency: deep discharge of a battery under low temperature conditions results in a rapid drop in energy efficiency; while as at higher temperatures, energy efficiency is generally not sensitive to deep discharge towards low cutoff voltages.

The fact that energy efficiency does not have a significant memory effect can also inspire us when designing battery management algorithms for BESS. For example, we can be more tolerant to short-term in-efficient operations, such as large-current and deep discharging in cold environment, for emergency situations without worries of long-term damage on energy efficiency, as long as we can resume favorable conditions after the situation is resolved. The possibility of individually or selectively controlling the charging and discharging of each battery based on these findings could be a way to maintain relatively high overall energy efficiency under dense energy throughput. This gives us

more flexibility to handle demand surges or deal with circumstances in the grid or the solar or wind farms that undergo unusual situations. Of course, because our data is much limited, whether extreme intensive use of the batteries under a lengthy period of unfavorable conditions will do irreversible harm to the energy efficiency still needs further investigations with additional experiments and data. This is also part of our future work.

5. Conclusions

Efficiency of batteries, particularly those used in ESSs, will have a significant impact on power systems. In this study, we proposed energy efficiency as an indicator of the battery's performance, and evaluated the energy efficiency of NCA lithium-ion batteries in the well-known dataset.

Our study examined the energy efficiency trends of these batteries under a variety of operating conditions. On the basis of the test of linear trend, we developed a energy efficiency trend model that is applicable to this dataset. We also discussed how different operating conditions under different scenarios affected the energy efficiency of the batteries. Results of the regression of the energy efficiency trend model have shown that increasing ambient temperature and decreasing discharge current have a positive impact on energy efficiency. The results also indicated that at low ambient temperatures, the battery's energy efficiency may be significantly reduced when operating at an extremely low cutoff voltage. As both aging and operating conditions have an impact on energy efficiency, BMS controllers should monitor the parameters of each battery, including terminal voltage, ambient temperature, charging and discharge current, so as to ensure performance for energy efficiency.

There are a number of limitations to this study that should be acknowledged. The size of the dataset currently used restricts further analysis of the impact of a wide range of operating conditions on energy efficiency trends. As our analysis is based on NCA lithium-ion batteries, it may be necessary to develop more complex models to estimate the energy efficiency of different lithium-ion batteries under a broader range of scenarios.

CRedit authorship contribution statement

Zihui Lin: Conceptualization, Validation, Visualization, Writing – original draft. **Dagang Li:** Validation, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Yuntao Zou:** Investigation, Resources, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data used in this study are publicly available. The dataset was sourced from the following repository: NASA dataset: Available at <http://ti.arc.nasa.gov/tech/dash/pcoe/prognostic-data-repository/#battery>.

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